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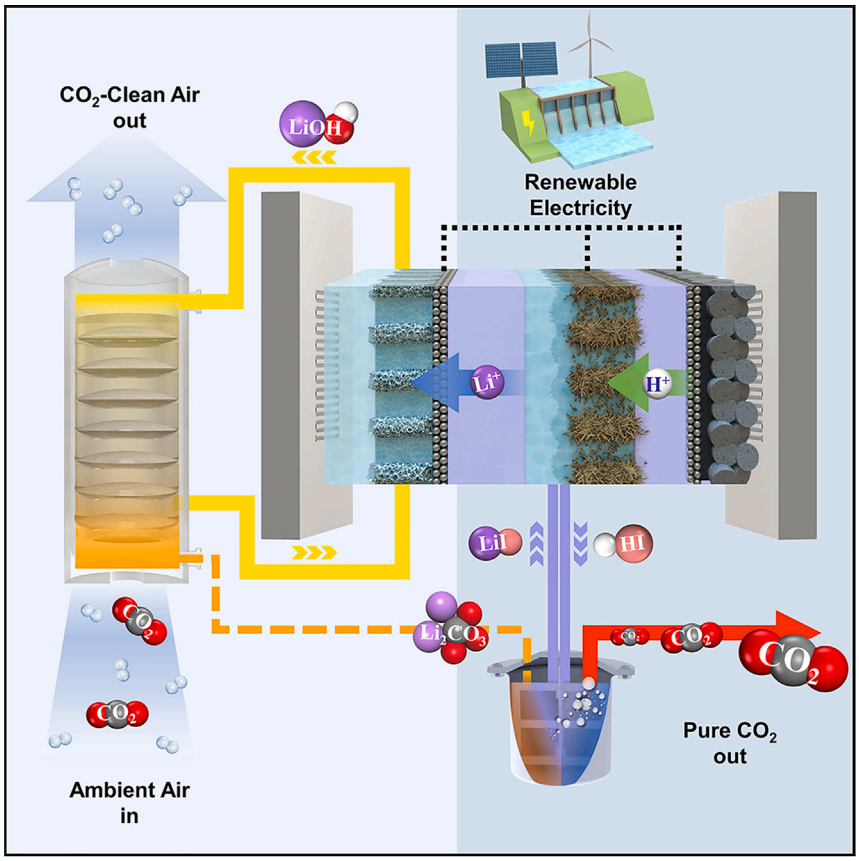

Direct air capture of CO_2 is expected to play a crucial role in meeting net-zero CO_2 emission goals. Current approaches are energy intensive and have a high associated emission intensity. We present an electrochemical pH swing DAC approach that can be powered by renewable electricity. We develop an alternating electrolyzer-fuel cell system that colocates redox reactions, on a shared common electrode, to efficiently regenerate alkaline liquids for CO_2 capture and acidic solutions for CO_2 liberation (6.4 GJ per tonne of CO_2 captured).

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Highlights

Electrocatalytic pH swing regenerates alkaline liquids for direct air CO_2 capture

Colocated redox reactions retain intermediates and lower cycle energy

Alternating electrolyzer-fuel cell operation is stable for over 100 h

An energy input of 6.4 GJ and an emission intensity of ~11 kg CO_2e per tonne CO_2 captured

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Regeneration of direct air CO_2 capture liquid via alternating electrocatalysis

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SUMMARY

The direct air capture (DAC) of carbon dioxide (CO_2) can potentially contribute to mitigating past and offsetting hard-to-abate future emissions; however, the regeneration of DAC capture liquids requires high temperatures and thermal energy inputs with emissions that diminish their net environmental benefit. Here, we present a low-temperature electrochemical process to regenerate alkaline capture liquids via alternating electrocatalysis (AE). Colocating oxidation and reduction reactions on a single electrode, cycled between electrolyzer and fuel cell modes, mitigates film formation and losses in the regeneration of alkali hydroxide and hydrogen halide. CO_2 can be captured and released with an energy input of 6.4 GJ/tCO_2 at 100 mA cm^{-2} and an emission intensity of ~11 kg CO_2e/tCO_2.

INTRODUCTION

Achieving net-zero carbon dioxide (CO_2) emission goals is expected to require the removal of CO_2 from the atmosphere in addition to broad decarbonization efforts.[1–4] Sectors such as aviation and shipping are particularly challenging to decarbonize.[5,6] Direct air capture (DAC) systems, coupled with CO_2 storage or with an upgrade into long-lasting materials, remove CO_2 directly from the atmosphere, thereby mitigating past emissions and compensating for persistent sources.[7,8]

DAC methods include alkaline capture liquids, typically metal hydroxides, and amine-based solid adsorbents to capture CO_2.[9–12] Captured CO_2 is liberated using thermal energy, simultaneously producing a stream of high-purity CO_2 and regenerating the capture media. This thermal energy is the dominant energy input to conventional DAC methods (~80% thermal and 20% electrical energies).[13] In the thermal regeneration process based on calcination, high-quality heat at ~900°C is required to regenerate the capture liquid.[14] These units are powered by fossil fuels, and the associated CO_2 emissions are substantial (~400 kg of CO_2 emitted per tonne captured, Figure 1A).[14] The high carbon intensity of conventional DAC methods motivates low-temperature DAC, ideally electrified.

The electrocatalytic regeneration of post-capture solutions is thus of potential interest.[15] The post-capture solution, in the form of an alkali carbonate electrolyte, can be fed into an electrolyzer where the carbonate anion and metal cation separate and electrochemically migrate toward the acidic and the alkaline compartments, respectively. The CO_2 is liberated in the acidic compartment using proton-induced pH swing, and the capture liquid is regenerated in the alkaline compartment

CONTEXT & SCALE

Large-scale direct air capture of CO_2 is a critical element of many net-zero emission plans. We present a means of electrochemical direct air capture (eDAC), compatible with renewable energy sources, to efficiently regenerate alkaline capture liquids via pH swing. Alkaline capture liquids are tolerant of oxygen and other impurities found in air and are capable of long-term industrial operation without chemical degradation. The isolation of the CO_2 capture and CO_2 liberation fluid loops in this eDAC approach yields a high-purity CO_2 stream, suitable for subsequent carbon sequestration or utilization. This eDAC approach can capture CO_2 with a low energy requirement and a significantly reduced emission intensity compared with conventional direct air capture methods, thereby improving the net environmental benefit. eDAC approaches allow for remote carbon removal schemes by reducing energy transmission infrastructure requirements and utilizing stranded renewable energy resources.

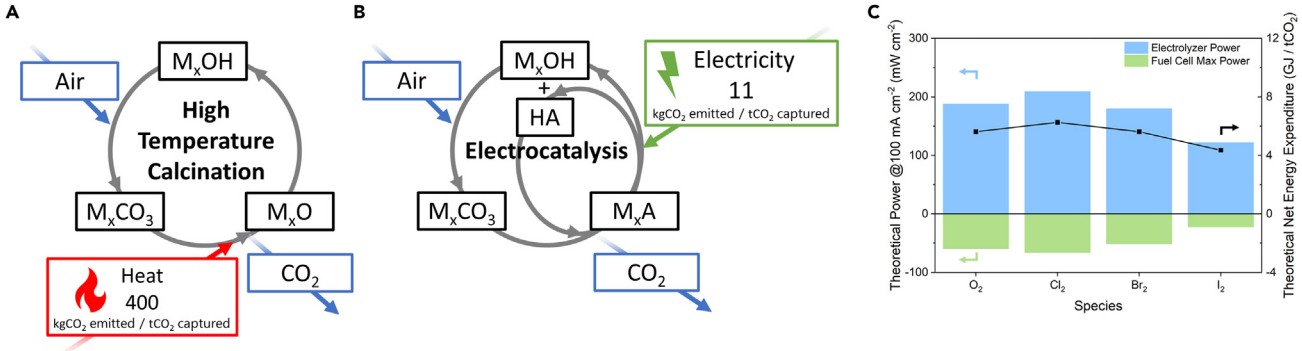

Figure 1. A method to regenerate post-capture solutions with low energy expenditure and carbon intensity

(A and B) Simplified process schematic of post-capture solution regeneration via (A) carbon-intense high temperature calcination and (B) electrocatalysis powered by renewable electricity.

(C) Estimated regeneration energy savings for the electrolyzer-fuel cell tandem by incorporating I_2 instead of other intermediates at 100 mA cm^{-2}.

through electrochemical hydroxide generation. Reactor configurations have been demonstrated based on a single cation exchange membrane (CEM),[16,17] double CEM,[18] and bipolar membrane (BPM).[19,20] Despite recent efforts to improve the efficiency of capture liquid regeneration and CO_2 liberation, the overall energy requirement of electrochemical DAC systems remains higher than that in thermal approaches (Table S1).

If the post-capture solution was chemically neutralized with acid, a tandem electrolyzer-fuel cell pair could regenerate the associated salt solution (Figure 1B). The electrolyzer regenerates alkaline capture liquids, whereas the fuel cell regenerates acidic solutions for CO_2 liberation.[21] To date, such approaches have relied on the oxygen (O_2) evolution reaction (OER) for proton generation. OER is a four-electron process and has much slower kinetics than the oxidation of halides, which are two-electron processes also capable of forming strong acids for CO_2 liberation, such as hydrochloric, hydrobromic, or hydroiodic acid (HI).[22] Among halogens, an iodine (I_2)-based system would require the lowest electrolyzer power (Figure 1C; supplemental information section theoretical net energy expenditure model of oxygen and halogen evolution/reduction reaction). We performed preliminary tandem electrolyzer-fuel cell experiments with water, chloride (Cl^-), bromide (Br^-), or iodide (I^-) anions and found that the oxidation of I^- exhibited the lowest overpotential, agreeing with theory (Figure S1). I_2 has additional advantages owing to its low volatility, mitigating the interference with captured gas phase CO_2 stream.[23,24] However, other reports of electrocatalytic I^- oxidation (e.g., in redox flow batteries) have demonstrated that continuous operation forms dense I_2 layers, which impose large charge transfer resistances and diminish the energy efficiency attainable with I_2 chemistry.[25–27]

Here, we present an electrocatalytic strategy that, by colocating cyclic I_2 redox reactions on a single common electrode, enables efficient regeneration of DAC capture liquid (Figure 2A). The lithium iodide (LiI) electrolyte combines I^- with a lithium metal cation. We employed lithium here owing to its single valence, small ionic radius, and low carbonate solubility. Neutralizing lithium carbonate (Li_2CO_3) precipitate with HI creates LiI electrolyte. The electrolyzer converts LiI electrolyte into lithium hydroxide (LiOH) capture liquid and I_2. The fuel cell then electrochemically hydrogenates I_2—a process that otherwise requires thermocatalytic hydrogenation at elevated temperature and pressure.[28] Hydrogen (H_2) evolved at the electrolyzer cathode is oxidized at the fuel cell anode, generating electrical energy to offset the electrolysis energy

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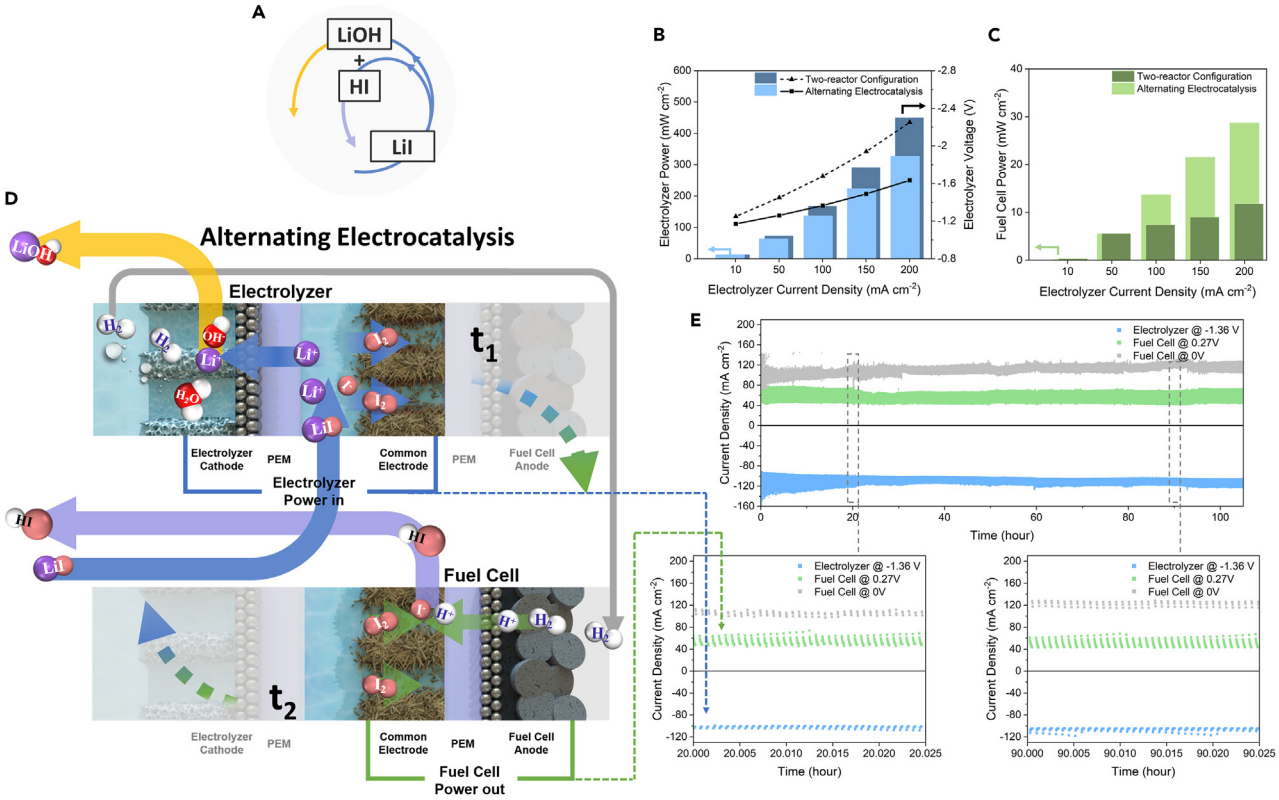

Figure 2. An alternating electrocatalytic process to efficiently react I_2 during capture liquid regeneration

(A) Process of alternating electrocatalysis (AE) splitting LiI electrolyte into LiOH capture liquid and HI acid.

(B) Reduced power input of AE relative to two-reactor configuration at different current densities.

(C) Increased power output of AE relative to two-reactor configuration at different electrolyzer current densities.

(D) Schematic of tri-electrode AE reactor cycling between electrolyzer and fuel cell operation.

(E) AE stability during prolonged operation.

demand. Both H_2 reactions have very fast kinetics, minimizing the net energy input.[29,30]

RESULTS

Capture liquid regeneration via I_2 electrosynthesis

We performed continuous I_2 electrosynthesis in a two-reactor tandem configuration (e.g., a dedicated electrolyzer and a fuel cell, Figures S2 and S3A, detailed preparation methods are shown in experimental procedures). When operated at current densities between 10 and 200 mA cm^{-2}, the voltage required by the electrolyzer increased substantially with current density (Figure 2B). At 200 mA cm^{-2}, the cell voltage reached −2.25 V with a high corresponding power input of 450 mW cm^{-2}, which was ∼1.7× the thermodynamic voltage (supplemental information section theoretical net energy expenditure model of oxygen and halogen evolution/reduction reaction). Some of the evolved I_2 accumulated on the anode, lowering the ionic conductivity of the system and further increasing the voltage over time (Figure S4).

Both half-reactions of the fuel cell (i.e., I_2 reduction and H_2 oxidation) have reactant availability impacted by the electrolyzer I_2/H_2 production rate. Therefore, at a constant electrolyzer current, we varied the fuel cell current (Figure S5) to identify the maximum fuel cell power output (i.e., product of current and voltage) (Figure 2C). The continuous fuel cell power output (7.3 mW cm^{-2} at 100 mA cm^{-2}) was much

lower than expected (>20 mW cm^{-2}, Figure 1C). At peak power output, the fuel cell voltage was close to the expected value, but the current was low (Table S2; Figure S5). The I^- present in the electrolyte and I_2 produced from the electrolyzer react chemically to form polyiodides.[31,32] In the reducing environment of the fuel cell cathode, these polyiodides (e.g., I_3^- or I_5^-) behave similarly to I_2 and can be electrochemically reduced directly into I^-.[32,33] The low fuel cell current suggested that the local I_2/polyiodide concentration at the fuel cell cathode was limiting. We concluded that conventional electrochemical approaches with independent electrolyzers and fuel cells cannot realize energy-efficient regeneration. I_2 film formation in the electrolyzer and a lack of the local availability of the I_2 redox intermediate in the fuel cell fundamentally limit the efficiency and stability of continuous operation.

Alternating electrocatalysis (AE) to reduce net energy input

Recognizing the challenge of locally retaining the iodide intermediate, we developed an alternating electrocatalysis (AE) approach. When the system is cycled between electrolyzer (t_1, Figure 2D) and fuel cell mode (t_2, Figure 2D), I_2 accumulated on a common electrode during electrolyzer operation is immediately available for conversion into HI when the system polarity is switched to fuel cell mode (Figures 2D and S3B). This approach could enhance energetic efficiency by limiting I_2 film growth to boost electrolyzer operation while limiting I_2 diffusion into the electrolyte to improve fuel cell operation. To perform I_2 electrosynthesis using an alternating approach, we developed a single tri-electrode reactor which, depending on the pair of electrodes activated, acts as an electrolyzer or a fuel cell. By incorporating a catalyst-coated membrane (CCM) and a gas diffusion electrode for the respective electrodes, the electrolyzer cathode and fuel cell anode were optimized for H_2 evolution and H_2 oxidation.

We tested the alternating approach using cyclic voltammetry with a range of frequencies between 0.005 and 1 Hz (Figure S6). At lower frequencies, the I_2 accumulated during oxidation was not fully recovered during reduction. As the I_2 film thickens, I_2 converts into polyiodides, diffuses into the bulk electrolyte, and lowers the local reactant concentration at the electrode. We found that frequencies higher than 0.5 Hz were sufficient to avoid excessive I_2 filming and diffusion, and further AE experiments were conducted with a frequency of 0.5 Hz (i.e., $t_1 = t_2 = 1$ s). We tested AE over a range of different frequencies and found that 0.5 Hz and above yielded comparable fuel cell power outputs (Figure S6E).

AE outperformed the two-reactor configuration as the electrolyzer and fuel cell both operated more efficiently. Unlike the two-reactor configuration, we operated the AE electrolyzer in potentiostatic mode to limit the voltage that the electrodes were exposed to while they were switched off (Figure S7). Linear interpolation was used to determine the electrolyzer voltage required to achieve the same time-averaged current densities as the two-reactor configuration (Figure S8). The AE electrolyzer voltages were much lower due to reduced film accumulation on the anode (Figure 2B). At 100 mA cm^{-2}, the electrolyzer in the two-reactor configuration required a cell voltage of −1.68 V, compared with −1.36 V for AE. The power required by the electrolyzer was likewise reduced: 168 mW cm^{-2} for the two-reactor configuration at 100 mA cm^{-2} vs. 136 mW cm^{-2} for AE at the same current (Figure 2B).

We analyzed the voltage distribution of an electrolyzer operating at 100 mA cm^{-2} (Figure S9). Nernst losses are the dominant source of voltage loss in the electrolyzer. Operating in the AE configuration, instead of the two-reactor configuration, significantly reduces I_2 filming, limits the I_2 concentration on the electrolyzer anode,

and reduces the Nernst loss (Figure S11; supplemental information section 1D COMSOL multiphysics model).

The fuel cell current was much higher for AE than in the two-reactor configuration at a given voltage due to improved I_2 availability on the electrode surface (Figure S12). AE yielded much higher fuel cell power outputs for all I_2/H_2 production rates (i.e., electrolyzer current densities) (Figure 2C). At a current density of 100 mA cm^{-2}, AE recovered 13.7 mW/cm^2, whereas the two-reactor configuration recovered only 7.3 mW cm^{-2}. The reduction in power input afforded by AE relative to the two-reactor configuration increased as the electrolyzer current increased.

We made pH measurements of the effluent to confirm that the AE electrolyzer and fuel cell can generate sufficiently alkaline and acidic solutions, respectively (Figure S13). After 34 min of operation, the regenerated LiOH capture liquid reached a pH of 13.1, and the HI acid reached a pH of 1.1. Since the pH change of the electrolytes was governed by electromigration through the membranes, the electrolytes reached similar pH values irrespective of the electrolyte flow rates (Figure S14).

AE stability

We demonstrated that AE operates continuously for 100 h with stable electrochemical performance and no apparent change in electrode morphology (Figures 2E and S15). We applied −1.36 V during electrolyzer mode, which maintained an average current density of 107 mA cm^{-2}. Within each cycle, the fuel cell operated at two-step voltages—initially at 0.27 V to yield the maximum power and then at 0 V to reduce any remaining I_2. Throughout the 100-h test experiment, the average current density during fuel cell mode was maintained at 51 mA cm^{-2} during the 0.27 V step and 110 mA cm^{-2} during the 0-V step. Analysis of the regenerated LiI electrolyte indicated that this system did not over-oxidize the I_2 into IO_3^- or IO_4^- (Figure S16).

Integrated DAC system based on AE

We generated a process flow diagram for a DAC system using AE to capture 1 kilotonne of CO_2 per year (Figure S17). In an integrated DAC system based on our regeneration system, atmospheric air would be driven into contact with LiOH, the alkaline capture liquid, forming solid Li_2CO_3 (Figure 3A). The Li_2CO_3 precipitates are neutralized with HI acid in the CO_2 liberator to produce a gaseous stream of pure CO_2 and aqueous LiI electrolyte (Figure 3B). Our experiments confirm that ~98% of the CO_2 captured by a LiOH capture liquid can be recovered in the CO_2 liberator (Figure S18; supplemental information section CO_2 liberator efficiency calculation). The LiI electrolyte generated in the CO_2 liberator is then used as the input for AE, where it is regenerated into capture liquid and acid for neutralization (Figure 3C).

By incorporating the electrolyzer energy inputs and the fuel cell energy outputs from our single-cycle experiments (Figure 2), we estimated the total energy expenditure per tonne of CO_2 captured from the air when implementing the AE regeneration strategy (Figure 4A). With a low energy requirement of 6.4 GJ/tCO_2 at 100 mA cm^{-2}, AE achieves a significant reduction in energy consumption when compared with the two-reactor configuration (Figure S19; supplemental information section whole system energy consumption and carbon intensity calculation). The energy requirement increased slightly with current density as all reaction overpotentials increased. The electrolysis energy was dominant, accounting for >95% of the net energy input at 100 mA cm^{-2}, whereas the fuel cell was able to recover 11% of the net energy input (Figure 4B). Although this total energy requirement is

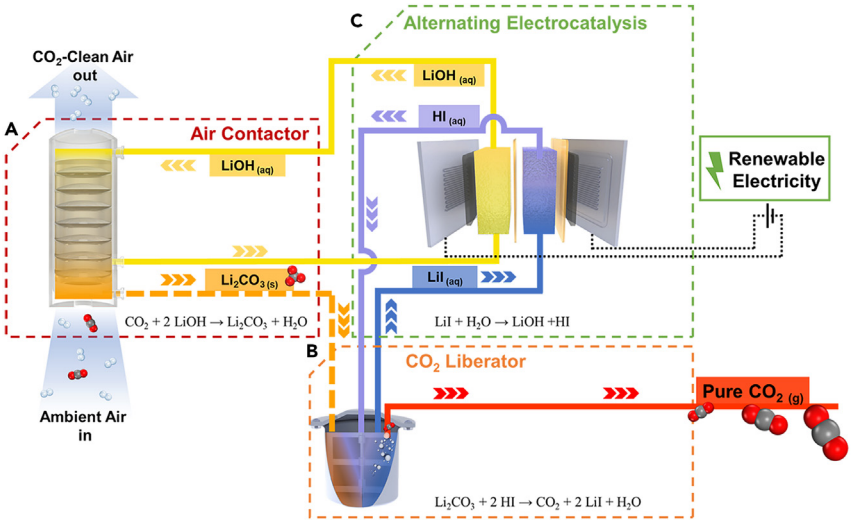

Figure 3. Complete process for DAC powered exclusively by electricity

(A) Air contactor for CO_2 removal from air via LiOH capture liquid. Solid Li_2CO_3 is formed in the process and sent to the liberator.

(B) CO_2 liberator wherein the concentrated Li_2CO_3 is neutralized with HI acid, producing LiI electrolyte and pure CO_2.

(C) AE wherein the LiI electrolyte is electrocatalytically regenerated into LiOH capture liquid and HI acid.

competitive with conventional DAC methods powered by thermal energy, the emission intensity is much lower (~11 kg CO_2e/tCO_2, supplemental information section whole system energy consumption and carbon intensity calculation).[10,14]

Demonstration of CO_2 capture from air with AE-regenerated liquids

AE-regenerated capture liquid can capture CO_2 from the air and release high-purity CO_2 when neutralized with the AE-produced HI (Figure S20). In scaled operation, the neutralization of Li_2CO_3 to CO_2 in the CO_2 liberator will eventually saturate the electrolyzer anolyte with CO_2 and introduce carbonic species into the electrolyte. Carbonic species can also be present in the electrolyzer catholyte, at levels below the Li_2CO_3 solubility limit, as a result of recirculation through the air contactor. To assess the impact on the electrolyzer, a carbon-containing catholyte (Li_2CO_3) was employed as the electrolyzer catholyte (in place of water), and carbonic species were introduced to the electrolyzer anolyte via a CO_2 purge. There were no side reactions detected between the carbonic species and the electrolyte (Figure S21). In this batch demonstration, the AE performance was the same with and without carbon-containing species in the electrolytes (Figure S22). These findings indicate the applicability of the AE regeneration system to post-capture solutions typical of DAC systems.

DISCUSSION

This work has demonstrated that AE can be used to efficiently regenerate inorganic alkaline capture liquids for DAC (Table S1; Figure S24). The AE system provided stable (100 h) operation without discernible electrocatalyst loss or damage when tested with a pair of large isolated electrolyte reservoirs (Figure 2). However, this lab testing does not fully reflect an integrated large-scale DAC operation (Figure 3). For example, O_2 content in the post-capture solution reduces energy efficiency. In the lab experiments, the post-capture solution was initially air-saturated (including O_2 content). Under these conditions, we found that O_2 reduction was limited to ~0.5% of the electrolyzer current density during the regeneration process

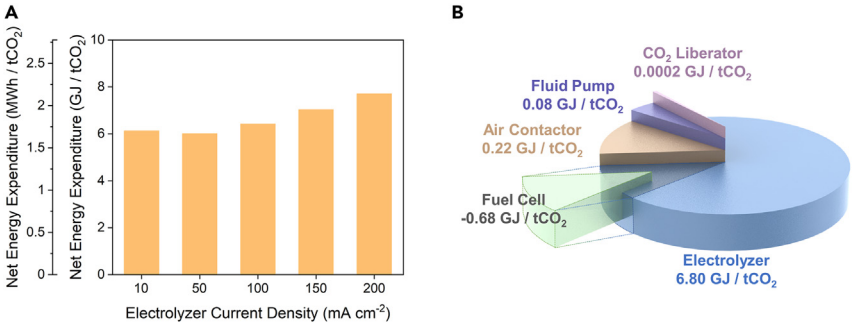

Figure 4. Performance of a DAC system incorporating AE for capture liquid regeneration
(A) Net energy expenditure for CO_2 capture at different current densities.
(B) Energy breakdown of the DAC system at 100 mA cm^{-2}. The fuel cell generates energy, offsetting the electrolyzer energy and reducing the energy input.

(supplemental information section impact of oxygen content in the capture solution). The effect of O_2 could be greater in a scaled contactor if there is an increase in the ratio of post-capture solution volume to electrolyzer surface area or if there is an increase in air pressure. Determining the cumulative effects of non-idealities in all components, and their impact on energy intensity and stability, will require larger scale testing of the trio of contactor, electrolyzer/fuel cell, and liberator.

With the AE approach, the Li_2CO_3 precipitates must be dried to prevent the transfer of LiOH capture solution to the CO_2 liberator, which would reduce the CO_2 liberator efficiency. The appropriate filtration technology for this step would depend on scale and other factors. The Solvay process provides a useful precedent here, with industrial production of sodium carbonate salt. Key to the Solvay process is a carbonation tower wherein an ammonium salt solution is exposed to CO_2 for the controlled precipitation of sodium bicarbonate. Typically, there are several towers to facilitate simultaneous washing and carbonation.[34] The precipitates are filtered using continuous processes, such as rotary vacuum filtration (which can reduce the water content to 1%–2%)[35] or centrifugation (which can reduce the water content to <0.05%).[36]

Conclusions

The electrochemical approach developed in this work employs design principles that are not constrained by conventional electrolyzers and fuel cell systems. The tri-electrode AE approach based on a halogen redox couple lowers the energy required for capture liquid regeneration. AE allows for I_2 to be generated at high local concentrations and then reacted rapidly into acid. When powered with renewable electricity, AE enables DAC with a simultaneous low energy cost and low carbon-intensity parameters that are especially critical given the scale on which CO_2 removal efforts are required.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, David Sinton (sinton@mie.utoronto.ca).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The data presented in this work are available from the corresponding authors upon reasonable request.

Catalyst and chemical preparation

All chemicals used for electrolytes were purchased from Sigma Aldrich (ACS reagent grade, >99.0%), including lithium chloride, lithium bromide, LiI, lithium sulfate, and sodium iodide. All electrolytes were prepared using Milli-Q grade deionized (DI) water (18.2 MΩ).

An iron nickel (FeNi) foam current collector for the electrolyzer cathode was purchased from Xiamen Lith Machine Limited. A CCM (Nafion 115 coated with platinum black [PtB], on one side) and hydrophobic carbon paper gas diffusion media (Sigracet 39 BB) were purchased from the Fuel Cell Store. The electrolyzer anode/fuel cell cathode (titanium felt coated with iridium mixed metal oxide [MMO] catalyst) was purchased from Magneto Special Anodes.

Two-reactor tandem membrane electrode assembly preparation

The Nafion 115 CCM (PtB, electrolyzer cathode) was sandwiched between the FeNi foam current collector and iridium-oxide-coated titanium felt (electrolyzer anode). Another Nafion 115 CCM (PtB, fuel cell anode) was sandwiched and hot-pressed between the hydrophobic carbon paper and iridium-oxide-coated titanium felt (fuel cell cathode) for the fuel cell (Figure S3A). The hot-press process was performed at a pressure of 7 MPa and a temperature of 220°F for 10 min.

Alternating electrocatalysis membrane electrode assembly preparation

The Nafion 115 CCM (PtB, electrolyzer cathode) and the FeNi foam current collector were assembled and compressed in the electrolyzer. The electrolyzer anode also acted as a fuel cell cathode (i.e., as an alternating common electrode). Another Nafion 115 CCM (PtB, fuel cell anode) was sandwiched and hot-pressed between the hydrophobic carbon paper and iridium-oxide-coated titanium felt (fuel cell cathode) for the fuel cell (Figure S3B). The hot-press process was performed at a pressure of 7 MPa and a temperature of 220°F for 10 min.

Electrocatalytic system operation

In the experiments, DI water was supplied to the electrolyzer compartment, whereas electrolyte (typically 3 M LiI) was supplied to the common compartment at ~10 mL/min. Both liquids were pumped into the compartments from the bottom, exhausted at the top of the compartment, and recirculated from a reservoir (10 mL for the electrolyzer catholyte and 50 mL for the LiI electrolyte) to the cell in a closed loop. For stability testing, an 800 mL reservoir was used for both electrolytes. H_2 gas from the electrolyzer compartment was fed directly into the fuel cell compartment (Figures S7 and S23). The dead volume in the electrolyzer cathode and fuel cell anode was minimized to ensure rapid transport of H_2 between the two compartments. The outlet of the fuel cell anode was submerged in water to retain H_2 and prevent air flow into the two compartments.

All electrochemical tests were performed using an electrochemical workstation (Autolab PGSTAT302N) operating at room temperature (~20°C) and atmospheric pressure (~1 atm). All reactors had an active area of 1 cm^2. During the two-reactor tandem configuration experiments, the chronopotentiometry and chronoamperometry measurements, performed by two independent workstations, were applied to the electrolyzer and fuel cell, respectively. A time-averaged current is reported

for chronoamperometry measurements. During the alternating electrocatalysis configuration experiments, the chronoamperometry measurements were made on the electrolyzer and fuel cell by two independent workstations with a frequency of 0.5 Hz ($t_1 = t_2 = 1$s). The electrolyzer and fuel cell circuits were isolated using an alternating microcontroller magnetic switch (Figure S7). During the 100 h experiment, the system alternated among chronoamperometry electrolyzer mode (−1.36 V, t_1 = 1 s), chronoamperometry fuel cell mode (0.27 V, t_2 = 1 s), and chronoamperometry fuel cell mode (0 V, t_0 = 0.5 s). Electrochemical data were recorded by NOVA Autolab software every 0.1 s. LiI and sodium iodide were evaluated in the alternating electrocatalysis configuration to test the compatibility of the AE system with various alkaline cations (Figure S24). For the cyclic voltammetry measurement, an H-cell configuration with an Ag/AgCl reference electrode was employed (Figure S25). All voltages reported are cell voltages without iR compensation.

Product analysis

The gas products collected from the CO_2 liberator were analyzed by gas chromatography (PerkinElmer Clarus 680). The regenerated capture solution was examined using a high-pressure ion chromatography system (Thermo Scientific Dionex Integrion HPIC). The HPIC eluent generator was set as 10 mM with 25 mA constant current mode for the suppressor to optimize the analysis of I^-, IO_3^-, and IO_4^- solutions (Figure S16). The HPIC eluent generator was set as 23 mM with 57 mA constant current mode for the suppressor to optimize the analysis of solution with and without carbonic species (Figure S22). The pump flow rate was 1 mL min^{-1} with a column pressure of 2,150 PSI and a column temperature of 30°C for all HPIC analyses. The pH of the regenerated capture liquid (LiOH) and acid (HI) were measured using pH meters (OMEGA PHH444 portable pH meter with Data Logging Function).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2023.07.011.

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AUTHOR CONTRIBUTIONS

D.S. and E.H.S. supervised the project. Y.X. and S.L. conceived the idea. Y.X. and S.L. designed and performed electrochemical experiments. Y.C.X., J.P.E., Y.Z., and R.K.M. assisted in electrochemical experiments. S.L., J.E.H., and M.F. performed carbon dioxide liberator experiments. Y.X. and S.L. performed product analysis. J.P.E., Y.C.X., Y.X., and S.L. analyzed the results and drafted the manuscript. All authors discussed the results and assisted during manuscript preparation.

DECLARATION OF INTERESTS

The authors declare the following competing financial interest(s): there is a patent application pending, filed by the authors of this article and their institutions.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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